Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Looping through lists and dataclasses will definitely help, a programming skill I hope to get better with is parsing through text files to find what I need

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The Easiest parts I believe will be looping through lists because I am already pretty good at that, The hardest I believe will be sorting the information to make it easily accessible through dataclasses since there will be a lot of information and im not sure if its all in the same order.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The tasks that will take the most time and effort will be making the drafter program

easy to use, and also the rating function to make a library the system can call back on. I believe sorting through all the food should be figured out first and then ratings should be focused on.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions would probably be wanting to play video games and watch sports with my friends. I will minimize them by setting up times where i am in other places doing my work so those distractions can't interfere.

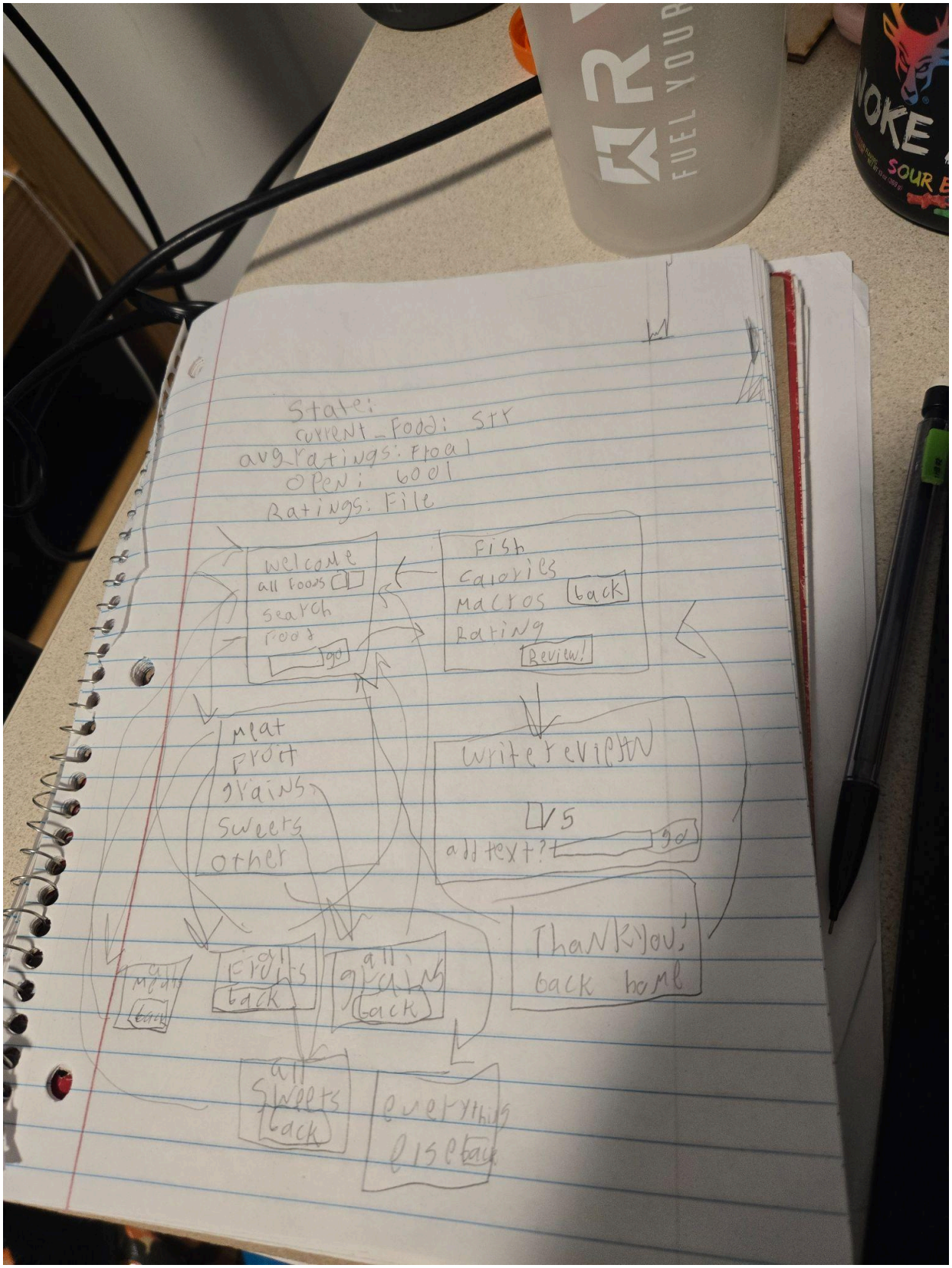
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


State:
current_food: str
avg_ratings: float
OPEN: bool
Ratings: File
welcome
all foods
search
food
go
fish
calories
macros
rating
back
Review!
meat
fruit
grains
sweets
other
write review
1/5
add text?
go
Thank you,
back home
meats
back
fruits
back
all
grains
back
all
sweets
back
everything
else
back

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	This will give users an easy way to access and view what the dining hall has to offer, see the nutritional value of it, and see what others are saying. Users are also able to add their own reviews, which will be kept in an outside library.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	current_food: str avg_ratings: float hall_open: bool ratings: file Current_food will have the name of the food the program is currently on. Avg_ratings will tell the user what other people have said on average in the past about this food Hall_open will tell the user if the dining hall that has it is open Ratings: is the file library that keeps all ratings of the selected food.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Food: Type: str Calories: int Carbs: int Protein: int Fat: int Vegetarian: bool Nothing here will need to be modified

Routes (minimum 7) The names of your route functions	Search Enter_Food Protein Grains Fruits Sauce Vegetables Dairy Beverages
Pages (minimum 5) A brief description of the pages in your site	Welcome page: asks you to type the food you are looking for or lets you choose a type of food youd like to see Food page: after searching shows up with the calories and average rating Rate page: asks users to enter a rating stored in library Food types page: shows multiple different food types to choose from if the user doesnt know what they want Sweets page: shows all foods that are registered as desserts
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox will be used to search for food Button will be used to move around the website in multiple different ways TextBox will also be used to give rating
If Statements (minimum 3) Brief description of what will be evaluated in your if statements	If user is looking for a certain type show them that type If user enters food that doesnt exist, show foods with

and how they will meaningfully change the logic of the site	what entered inside it, otherwise, show food does not exist If user leaves rating, the rating library is updated
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	Loops through the list takes from the text file and looks for any food that match or have the food listed inside of it.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM to assist me in this assignment

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.

set_website_title("Dining Hall Finder")

set_site_information(author="Logan Mattie",

description=

sources= "ChatGPT",

planning=".pdf"

links=[https://github.com/LoganM6281/Dining-Hall-Finder]